

SubSense

An AI-Enabled Wireless Surface Mesh Network for Real-Time Subsidence Monitoring
and Early Warning in Underground Coal Mines

Problem Statement, Proposed Solution and System Uniqueness

Domain: Smart Mining Safety | Focus Area: IoT, Wireless Mesh Networking, Artificial Intelligence, GIS

1. Problem Description & Context

Underground coal mining in India frequently causes surface subsidence – the gradual or sudden sinking of ground above worked-out mine panels. This poses serious risks to nearby habitations, roads, railway lines, agricultural land, forests, and mine infrastructure, and has historically led to structural damage, displacement, and in severe cases, loss of life.

Subsidence monitoring in Indian coalfields today relies largely on manual field surveys, periodic total-station or levelling measurements, and post-facto damage assessments carried out only after cracks or ground sinking become visible. This reactive approach has critical gaps: monitoring intervals are too infrequent to catch the onset of ground movement, surveys are labour-intensive and do not scale across the hundreds of active underground panels nationwide, and by the time subsidence is detected through conventional means, the window for preventive action – evacuation, load restriction, or reinforcement – has often already closed.

At the same time, low-cost IoT hardware (ESP32/Arduino-class microcontrollers), long-range low-power wireless communication (LoRa/Zigbee/Wi-Fi mesh), and AI/ML-based anomaly detection have matured enough to be deployed affordably at scale – yet Indian mines have no indigenous, real-time, sensor-network-based early warning system built on these technologies. Most existing subsidence-prediction research relies on satellite InSAR data, which offers wide-area coverage but suffers from days-to-weeks revisit times, cloud-cover limitations, and coarse spatial resolution – insufficient for panel-level, real-time early warning.

This creates a clear need for a low-cost, indigenous, real-time ground-level monitoring system: one that senses micro-scale ground deformation (tilt, vibration, displacement, crack initiation) continuously and directly above active mine panels, uses a self-healing wireless mesh network suited to remote and rugged mine terrain, and applies AI/ML to flag abnormal deformation patterns early enough to give mine operators and regulators actionable lead time – shifting subsidence management from reactive damage assessment to proactive risk mitigation, in line with India's smart and sustainable mining goals.

2. Proposed Solution & Workflow

SubSense is an AI-enabled smart mine subsidence monitoring and early warning platform built on a localized wireless surface mesh sensor network deployed directly above underground mine panels. The system continuously senses micro-scale ground deformation, correlates readings across nodes, and uses AI/ML to detect abnormal patterns and predict subsidence risk zones well before visible damage occurs. The workflow below integrates both the baseline system design and the additional uniqueness features described in Sections 4 and 5.

Step 1 - Sensing Layer: Distributed Smart Nodes

Low-cost nodes (ESP32/Arduino-based) are deployed in a grid across the surface above the mine panel, each equipped with tilt/inclination sensors, vibration sensors, displacement/stretch sensors, crack detection sensors, and optional low-cost GPS/positioning modules. Nodes are powered by solar and battery, supplemented by piezoelectric energy harvesting that converts ambient ground vibration into a trickle charge. High-risk nodes additionally carry a local siren/beacon that issues an immediate on-ground audible and visual warning before any alert reaches a phone or dashboard.

Step 2 - Communication Layer: Wireless Mesh Backbone

Nodes relay data via a self-healing LoRa-based mesh network, supplemented by Zigbee for dense sub-clusters, hopping data node-to-node toward a gateway. This avoids dependence on cellular coverage, tolerates individual node failure or damage, and keeps power draw low enough for multi-year battery/solar operation. Each node also reports its own battery level and signal strength, enabling predictive maintenance before a node actually fails.

Step 3 - Gateway, Edge Buffering & On-Device Intelligence

A gateway node aggregates mesh data, performs pre-filtering, and buffers readings locally during connectivity loss, syncing to the cloud once a Wi-Fi/4G backhaul link is available. Lightweight TinyML models run directly on the gateway (and where feasible, on individual nodes), allowing anomaly detection and alerting to continue even with zero internet connectivity, with model updates pushed from the cloud once reconnected.

Step 4 - AI/ML Intelligence Layer

Cloud-side and edge-capable models analyse live and historical multi-sensor time-series data to detect anomalies in individual and cross-node readings, correlate signals spatially using a graph-based (GNN-inspired) model over the mesh topology, estimate severity and progression rate, and generate a predictive time-to-critical countdown for each risk zone. Every alert is accompanied by a confidence score and a short explanation of which sensors and nodes triggered it, and operator feedback on false alarms is automatically fed back to refine detection thresholds over time.

Step 5 - GIS Visualization, Digital Twin & Risk Mapping

Detected anomalies and predictions are overlaid on a live GIS map of the mine panel, generating deformation heatmaps and risk-zone boundaries. A 3D digital twin of the panel combines subsurface mine-planning geometry with real-time surface deformation data, and periodic cross-validation against freely available satellite InSAR data (e.g. Sentinel-1) provides a coarse macro-scale sanity check on the fine-grained ground sensor predictions.

Step 6 - Alerting & Decision Support

When risk thresholds are crossed, the system automatically triggers severity-tiered alerts (advisory, warning, critical) via SMS, email, and mobile push notification to mine operators, safety officers, and regulators. A parallel community alert channel sends localized-language SMS and automated voice calls to residents registered in the affected zone, ensuring the warning reaches low-literacy or non-digital populations as well.

Step 7 - Dashboard, Reporting & Multi-Site Scaling

A web/mobile dashboard, built as a multi-tenant platform, provides real-time sensor status, historical trend charts, risk-zone maps, alert logs, a historical replay and what-if simulation mode for planners, and one-click export of DGMS-compliant regulatory reports. The multi-tenant design allows a single deployment to serve multiple mine sites and operators, directly supporting scale-up across Indian coalfields without architectural redesign.

3. Key System Requirements

The system is designed to be low cost, low power, scalable, offline-capable, and reliable under field conditions typical of remote Indian coalfields. Core requirements include:

- Continuous real-time sensing of tilt, vibration, displacement, and crack initiation.
- Multi-hop, self-healing wireless mesh communication with automatic route recovery.
- Local buffering during connectivity loss with periodic cloud synchronization.
- AI/ML-based anomaly detection, severity estimation, and progression prediction.
- GIS-based live deformation mapping and automated, severity-tiered alerting.
- Target mesh data delivery reliability above 95% under field conditions.
- Multi-month to multi-year node operating life on battery, solar, and harvested power.
- Architecture that scales from a single-panel pilot to multi-coalfield deployment.

4. Given Features - Baseline System

These are the core, foundational features that make up the baseline SubSense platform as originally proposed, organised by system layer.

Feature	Category	Description
Multi-parameter sensor node	Hardware	Single node integrates tilt, vibration, displacement/stretch, and crack detection sensors.
Optional GPS positioning	Hardware	Low-cost GPS module for absolute node position tracking.
Solar + battery power	Hardware	Standard field power supply for continuous unattended node operation.
Configurable sampling rate	Hardware/Software	Per-sensor sampling frequency can be tuned to balance power use and data resolution.
Self-healing LoRa mesh backbone	Hardware/Software	Long-range, low-power multi-hop network that reroutes automatically around failed nodes.
Zigbee sub-clustering	Hardware/Software	Optional denser local mesh for tightly spaced nodes around a specific risk zone.
Gateway with Wi-Fi/4G backhaul	Hardware	Aggregates mesh traffic and uplinks to the cloud where connectivity is available.
Unsupervised anomaly detection	Software	Isolation Forest / Autoencoder models flag abnormal sensor behaviour without labelled data.
Time-series progression forecasting	Software	LSTM-based models forecast how detected deformation is likely to evolve.
Spatial interpolation (Kriging)	Software	Generates a continuous risk surface across the panel from discrete node readings.
Cross-sensor, cross-node correlation	Software	Combines multiple sensor types and neighbouring nodes to reduce false positives.

Feature	Category	Description
Live GIS deformation heatmap	Software	Real-time visual map of deformation intensity and risk zones.
Tiered SMS/email/app alerts	Software	Automated notifications classified as advisory, warning, or critical.
Role-based dashboard	Software	Separate views and permissions for operators, planners, and regulators.
Time-series database	Software	InfluxDB/TimescaleDB storage optimised for high-frequency sensor data.
Offline edge buffering	Software	Local storage of readings during connectivity loss, synced once online.

5. Added Features - Uniqueness

The following features were added specifically to differentiate SubSense from generic IoT-plus-AI mining proposals. They are grouped by hardware and software category, and are positioned as the platform's core competitive uniqueness.

Feature	Category	Description
Piezoelectric energy harvesting	Hardware	Converts ambient ground vibration into supplementary charging power, so the phenomenon being monitored partially powers the sensor detecting it.
Digital twin of the mine panel	Software	Live 3D model combining subsurface mine-planning geometry with real-time surface deformation data for predictive visualization.
TinyML on-device inference	Software	Lightweight anomaly detection running directly on gateway/nodes, enabling true offline detection and alerting with periodic model updates from the cloud.
GNN / swarm-intelligence correlation	Software	Graph-based spatial correlation model treating the mesh as a graph of nodes and edges, improving detection of genuine subsidence patterns over generic threshold-based methods.
Explainable, confidence-scored alerts	Software	Every alert reports which sensors and nodes triggered it and a confidence score, addressing false-alarm and black-box AI concerns.
DGMS-compliant report export	Software	Auto-generated statutory-format reports for direct use in regulatory compliance workflows.
Mesh self-health & predictive maintenance	Hardware + Software	Nodes report battery and signal health; the system predicts node failure and schedules maintenance before data loss occurs.
Community/citizen SMS alert layer	Software	Extends alerting beyond operators to residents in the risk zone via localized-language SMS.
Multi-hazard extensibility	Hardware/Architecture	Modular sensor daughterboard design allows the same mesh and platform to later support gas, fire, or roof-fall sensing without redesign.
Onboard local siren/beacon	Hardware	High-risk-zone nodes carry a local audible/visual alarm, warning nearby workers instantly even before a remote alert is delivered.
Predictive time-to-critical countdown	Software	AI output includes an estimated time window before a zone reaches critical deformation, not just a binary alert.
Historical replay & simulation mode	Software	Dashboard mode to replay past events or simulate future extraction scenarios for planning and training.

Feature	Category	Description
Automatic false-alarm learning loop	Software	Operator feedback on false positives automatically retrains and adjusts detection thresholds without manual data-science effort.
Multi-tenant SaaS dashboard	Software	A single platform deployment can host and separate data for multiple mine sites and operators, supporting national-scale rollout.
Voice-based community alerts	Software	Automated local-language voice calls supplement SMS for low-literacy or non-digital populations near mine sites.
Satellite InSAR data fusion	Software	Periodic cross-validation of ground-sensor predictions against Sentinel-1 InSAR data as a coarse macro-scale hybrid check.

6. Expected Outcome & Impact

- **Safety:** Early, actionable warning shifts mine safety response from reactive damage assessment to proactive risk mitigation, reducing risk to life, workers, and property.
- **Economic:** Substantially lower monitoring cost than imported instrumentation or satellite InSAR subscriptions, reducing losses from unplanned subsidence damage.
- **Operational:** Continuous, auditable monitoring data supports safer panel design, extraction sequencing, and regulatory compliance reporting.
- **Environmental & Social:** Protects agricultural and forest land, and improves risk communication with nearby communities through direct, localized alerts.
- **Strategic:** Establishes an indigenous, exportable, Made-in-India monitoring technology stack, extensible to other underground hazard-monitoring use cases.

7. Conclusion

SubSense combines a distributed wireless surface mesh, low-cost multi-sensor nodes, and an AI/ML intelligence layer into a single indigenous platform for real-time underground coal mine subsidence monitoring. Beyond its baseline sensing and alerting capability, the platform's added uniqueness features – from energy-harvesting hardware and on-device TinyML inference to explainable AI, digital twins, and multi-tenant scaling – position it as a genuinely differentiated, deployable, and nationally scalable solution for smart and sustainable mining safety in India.